

### ✓ CS407 Final Term Important MCQs

1. Which of the following is a type of learning in AI?

A. Supervised

B. Unsupervised

C. Reinforcement

☐ D. All of the above ✓

2. In which AI technique, the machine learns from past actions and uses feedback?

A. Supervised Learning

☐ B. Reinforcement Learning ✓

C. Unsupervised Learning

D. Regression

3. Which search algorithm uses a queue data structure (FIFO)?

☐ A. Breadth First Search (BFS) ✓

B. Depth First Search (DFS)

C. Best First Search

D. A\* Search

4. A algorithm is a combination of:\*

A. Uniform Cost + DFS

☐ B. Uniform Cost + Heuristic ✓

C. BFS + DFS

D. Greedy + DFS

5. Which of the following is a knowledge representation technique?

A. Semantic Network

B. Frames

C. Rules

☐ D. All of the above ✓

6. What is the branching factor in BFS if each node has 3 children?

☐ A. 3 ✓

B. 6

C. 1

D. None

7. Which AI paradigm focuses on mimicking human thought processes?

☐ A. Cognitive AI ✓

B. Classical AI

C. Reactive AI

D. Probabilistic AI

8. Which logic is used in Prolog?

A. Predicate Logic

☐ B. First-Order Logic ✓

C. Fuzzy Logic

D. Modal Logic

9. Which of the following is not a property of heuristic function?

A. It estimates cost to goal

☐ B. It guarantees the shortest path ✓

C. It's problem specific

D. It's used in A\*

10. Which programming language is closely associated with AI?

A. Python

B. Lisp

C. Prolog

☐ D. All of the above ✓

11. What is the output of an inference engine?

A. Query

☐ B. Conclusion ✓

C. Rule

D. Dataset

12. Which algorithm backtracks when stuck?

☐ A. DFS ✓

B. BFS

C. A\*

D. UCS

13. Which data structure is used in DFS?

☐ A. Stack ✓

B. Queue

C. Linked List

D. Priority Queue

14. Which search is guaranteed to find the shortest path?

A. DFS

☐ B. BFS ✓

C. Hill Climbing

D. Greedy

15. In AI, what does PEAS stand for?

☐ A. Performance measure, Environment, Actuators, Sensors ✓

B. Plan, Execute, Analyze, Solve

C. Problem, Environment, Algorithm, Solution

D. None

16. Which of these is a machine learning technique?

A. Decision Tree

B. Neural Networks

C. Support Vector Machine

☐ D. All of the above ✓

17. Which algorithm suffers from local maxima?

A. BFS

☐ B. Hill Climbing ✓

C. A\*

D. DFS

18. Which one is a blind search?

A. Best-First Search

☐ B. BFS ✓

C. A\*

D. Hill Climbing

19. Which logic allows truth values between 0 and 1?

☐ A. Fuzzy Logic ✓

B. Binary Logic

C. Modal Logic

## D. Propositional Logic

20. The Turing Test is used to test:

☐ A. Intelligence of a machine ✓

B. Accuracy of algorithm

C. Performance of robot

D. None

21. Which of these is a part of Natural Language Processing?

A. Syntax Analysis

B. Semantic Analysis

C. Pragmatics

☐ D. All of the above ✓

22. In neural networks, weights are used to:

☐ A. Adjust the influence of input ✓

B. Normalize data

C. Generate output

D. Store bias

23. Genetic Algorithms are inspired by:

A. Physics

☐ B. Biology ✓

C. Chemistry

D. Robotics

24. Which of these is used for face recognition?

☐ A. Deep Learning ✓

B. DFS

C. BFS

D. Alpha-Beta Pruning

25. What is the main goal of AI?

A. Create databases

B. Solve mathematical problems

☐ C. Make machines intelligent ✓

D. Increase memory usage

26. What is pruning in search trees?

☐ A. Removing unnecessary nodes ✓

B. Adding more children

C. Updating heuristic

D. None

27. Which of the following is NOT a learning method?

☐ A. Forgetting ✓

- B. Supervised Learning
- C. Reinforcement Learning
- D. Unsupervised Learning

28. Which algorithm is optimal and complete?

- A. DFS
- B. Hill Climbing
- ☒ C. A\* ✓
- D. None

29. Which one uses fitness function?

- ☒ A. Genetic Algorithm ✓
- B. A\*
- C. Hill Climbing
- D. Decision Tree

30. Knowledge base in an expert system contains:

- ☒ A. Facts and Rules ✓
- B. Only facts
- C. Data
- D. Queries

31. Which search algorithm is best for small depth solutions?

☐ A. BFS ✓

B. DFS

C. Hill Climbing

D. GA

32. Which of the following is a feature of intelligent agents?

A. Autonomy

B. Perception

C. Rationality

☐ D. All of the above ✓

33. An agent perceives its environment through:

A. Sensors ✓

B. Effectors

C. Actuators

D. Functions

34. Which is the main component of a neural network?

☐ A. Neuron ✓

B. Token

C. Frame

D. Fact

35. Which AI field includes voice recognition?

- ☒ A. NLP ✓
- B. Planning
- C. Vision
- D. Game Playing

36. Which rule-based system uses IF-THEN rules?

- ☒ A. Expert System ✓
- B. A\*
- C. NLP
- D. Neural Net

37. What is an ontology in AI?

- ☒ A. Formal representation of knowledge ✓
- B. A database
- C. A language
- D. A goal function

38. Which is NOT an AI application?

- A. Robotics
- ☒ B. MS Word ✓
- C. Game Playing
- D. Face Recognition

39. Heuristic function is also known as:

☐ A.  $h(n)$  ✓

B.  $g(n)$

C.  $f(n)$

D.  $\text{cost}(n)$

40. Which learning uses labeled data?

☐ A. Supervised ✓

B. Unsupervised

C. Reinforcement

D. Deep

41. Which algorithm selects node with lowest evaluation function  $f(n)$ ?

☐ A.  $A^*$  ✓

B. DFS

C. Hill Climbing

D. UCS

42. Which of these is used in robotics for motion planning?

☐ A.  $A^*$  ✓

B. BFS

C. IDDFS

D. Alpha-Beta

43. The process of deriving new facts from existing facts is called:

☐ A. Inference ✓

B. Induction

C. Deduction

D. Propagation